

Kaggriculture — Bug & Defect Register

Current repository review. This report converts the identified bug/defect list into a structured PDF. Severity reflects downstream runtime, economic, strategic, statistical, or validation impact.

Severity	#	Bug / Defect	Impact / Description	Domain
P0	1	Animal economy not end-to-end enabled	Animal production path is not fully operational end-to-end. This can leave intended animal economics unavailable or strategically ineffective.	Strategic/economic
P1	2	Goose/coop path is unreachable	The goose/coop lifecycle is not sufficiently connected to the active strategy/economy path.	Functional/strategic
P0	3	Wheat/animal bootstrap dependency unresolved	The dependency between wheat production/resources and animal bootstrap can prevent the intended economic path from becoming self-starting.	Economic/runtime
P0	4	Planning state can diverge from emitted market orders	Planner decisions can be made from state that does not exactly reflect orders actually emitted, creating stale inventory/cash/order assumptions.	Economic/state
P1	5	Market-order priority is not globally optimized	SELL/HIRE/generated BUY orders are arbitrated in stages rather than by a single globally optimized economic objective.	Strategic/economic
P1	6	New animal purchases are not fully included in feed reservation	Feed planning does not fully reserve resources for animals purchased in the same decision cycle.	Economic/functional
P1	7	Pasture capacity is calculated independently by multiple components	Multiple components can maintain their own capacity assumptions, creating potential drift in resource availability and animal counts.	Architecture/state
P1	8	Individual carried-animal state is not modelled correctly	The lifecycle of a specific animal while carried/being placed is not represented with sufficient state fidelity.	Functional/state
P1	9	Animal pickup/placement is not atomic	Animal acquisition and placement are split across turns without an atomic lifecycle guarantee.	Runtime/functional
P1	10	'taken' is overloaded across operation types	A shared state concept is used for materially different operation/resource semantics, increasing collision and accounting risk.	Architecture/state
P1	11	HARVEST/FERTILIZE collision risk	Harvest and fertilizer-related operations can compete for the same worker/resource path without a sufficiently explicit reservation model.	Functional/economic
P1	12	Planting intent is calculated more than once	Repeated planning of planting can create inconsistent intent, especially when state changes between planning stages.	State/strategy
P1	13	Wheat planting competes with mandatory infrastructure construction	The scheduler does not sufficiently arbitrate wheat planting against infrastructure needed to unlock the broader economy.	Strategic/economic
P1	14	Wheat shed pickup lacks complete per-turn resource reservation	Resource pickup is not fully reserved across the turn, allowing planning assumptions to diverge from actual availability.	Runtime/state
P1	15	Worker local-state fallback can become stale	Dead-reckoned worker state can persist after the real game state changes, causing incorrect routing or action selection.	Runtime/functional
P1	16	Exception containment hides operational failure	Broad exception handling converts failures into safe-looking output while potentially concealing the underlying defect from the game/telemetry path.	Runtime/observability
P2	17	Telemetry persistence is not a reliable validation mechanism	Telemetry exists but does not itself establish that all important runtime/game-state failures have been captured and validated.	Observability
P2	18	Market/shop information has multiple sources	Duplicated market/shop knowledge increases the risk of rule drift between components.	Architecture
P2	19	Market demand approximation is too close to engine facts	Strategy-level demand logic risks duplicating game-domain behavior rather than treating the validated rules artifact as authoritative.	Architecture
P2	20	Crop timing has too many independent temporal gates	Multiple timing cutoffs can interact and produce unexpected planting/harvest decisions.	Strategy/functional
P2	21	Care capacity is advisory rather than a hard resource constraint	The scheduler can plan acreage without a complete hard constraint tying crop count to actual care capacity and worker throughput.	Economic/functional

Severity	#	Bug / Defect	Impact / Description	Domain
P2	22	Hungarian matching optimizes distance, not total economic utility	Worker-task assignment minimizes travel distance but does not directly optimize task value, urgency, opportunity cost, or economic impact.	Strategic
P2	23	Priority bands can conceal materially different risk	Tasks in the same band may have substantially different consequences if delayed, reducing scheduling precision.	Strategy
P2	24	Generic 'leave a day to sell' crop cutoff	A generic terminal cutoff may unnecessarily constrain crops or fail to reflect exact engine timing and market conditions.	Strategy/economic
P2	25	Unplayed matchup returns 0.5 in empirical BT reporting	An unobserved matchup is represented as an even empirical probability, which can be mistaken for evidence rather than lack of data.	Statistical
P1	26	Zero-win BT agents receive an ad-hoc 1e-6 strength	The fitted strength floor is not a principled uncertainty/regularization treatment and can create extreme rankings under sparse data.	Statistical
P1	27	Disconnected BT comparison graphs are not handled	Bradley–Terry inference across disconnected matchup components is not identifiable as one coherent global ranking.	Statistical
P2	28	No uncertainty interval for BT latent strengths	The model reports point strengths without a confidence/credible interval for latent ability estimates.	Statistical
P1	29	No held-out BT validation	There is no demonstrated held-out evaluation showing that the BT model generalizes beyond the matches used for fitting.	Validation/statistical
P1	30	No demonstrated exogenous opponent pool	There is no demonstrated broad, independent opponent set sufficient to establish robustness against diverse strategies.	Validation/strategic
P0	31	Unit-test success is insufficient for exact-engine correctness	Passing the current unit suite does not establish correctness against the actual competition engine and its complete action/state semantics.	Validation
P0	32	No demonstrated exact-engine multi-opponent validation	There is no demonstrated evidence from exact-engine runs against multiple opponents sufficient to establish competitive correctness.	Validation
P1	33	No demonstrated overfitting/underfitting evidence	There is no demonstrated held-out/paired evidence sufficient to conclude that strategy tuning is neither overfit nor underfit.	Validation/statistical
P1	34	\$35,000 threshold is not a universal release criterion	The documented threshold was calibrated against a limited baseline and should not be treated as a universal competition-readiness gate.	Validation
P1	35	Animal capability documentation exceeds measured reliability	Documented capability can exceed what has been demonstrated by broad empirical validation, creating a risk of overstated readiness.	Documentation/validation

Severity Summary

P0 — defects that can block core functionality, cause material economic/runtime failure, or invalidate correctness claims.

P1 — significant functional, strategic, architectural, statistical, or validation defects that can materially degrade behavior or confidence.

P2 — lower-severity but still substantive defects, limitations, or design/validation gaps.

Severity	Count	Primary concern
P0	5	Core functionality, state/economic correctness, exact-engine validation
P1	19	Functional/strategic defects, BT inference limitations, validation gaps
P2	11	Architecture, strategy precision, observability and statistical reporting

Validation note

The register distinguishes code-level defects from statistical and validation limitations. Passing the existing unit-test suite should not be interpreted as proof of exact-engine correctness, multi-opponent robustness, Bradley–Terry robustness, or competition readiness.